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/*****
*   COMPANY                      : TAKEDA
*   PROTOCOL/STUDY/PROJECT      : ShortCourse
*   AUTHOR                      : OS Adjusted Analysis Working Group
*   DATE                        : 08Sep2025
*   PROGRAM NAME                : macro_km_plot.sas
*   LOCATION                    : /.../.../...
*   VERSION/ PLATFORM          : SAS V9.4 / LINUX
*   PURPOSE                     : macro to plot adjusted KM
*   MACROS CALLED                :
*   INPUT                       :
*   OUTPUT                      :
*   PARAMETERS (Optional)       : INMAIN:      Name of final weighted analysis data
*                                : INSTRATDS:   Name of final statistic data contains stratified p-value, HR and 95% CI
*                                : INUNSTRATDS:  Name of final statistic data contains unstratified p-value
*                                : METHOD:        Weighted method
*                                : XOVER:        Xover Y/N (default is N)
*                                :
*   NOTES (Optional)            : ASA Traveling Course, September to October 2025
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%macro macro_km_plot(inmain=,instratds=,inunstratds=,method=,xover=N);
ods exclude all;

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%if &xover=Y and &method=MSM %then %do;
data base;
    id=1;
    crossover=0;
    output;
run;

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data ads;
    set &inmain;
    by usubjid;
    id=1;
run;

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data ads;
    merge ads(in=b) base(in=c);
    by id;

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        if xover ne crossover then
            stabw=0;
run;
%end;

%else %do;
data ads;
    set &inmain;
run;
%end;

*** reorganize data for proc phreg;
*** set the ruler;

proc sort data=ads out=temp;
    by usubjid;
run;

proc univariate data=temp noprint;
    var analdy;
    output out=tp max=maxday;
run;

data tp1;
    set tp;
    maxday=round(maxday);
run;

data _null_;
    set tp1;
    call symput('maxd', put(maxday,best.));
run;

data temp;
    set temp;
    by usubjid;

    if last.usubjid;
    keep usubjid;
run;

data temp;

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        set temp;
        by usubjid;
        array a{&maxd.};

        do i=1 to &maxd.;
            a(i)=i;
        end;
        drop i;
run;

*** prepare data for KM plot;
*** first, set correct count for atRisk;

proc transpose data=temp out=new1;
    by usubjid;
run;

data new1;
    set new1;
    drop _name_;
    rename col1=analdy;
run;

proc sort data=new1 out=new1a;
    by usubjid descending analdy;
run;

proc sort data=ads out=new2a;
    by usubjid descending analdy;
run;

data new2;
    merge new1a new2a;
    by usubjid descending analdy;
    day=round(1*analdy);
    keep usubjid die stabw trt day;
run;

data new2;
    set new2;
    by usubjid descending day;
    retain wt rx;

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        if first.usubjid and stabw=. then
            wt=-1;
        else
            do;

                if stabw ne . then
                    wt=stabw;

                if trt ne . then
                    rx=trt;

            end;
run;

data new2;
    set new2;
    by usubjid descending day;

    if wt=-1 then
        delete;

    if die=. then
        die=0;
    wtdie=die*wt;
run;

proc sort data=new2;
    by usubjid day %if &xover=N %then %do; wt %end; die;
run;

data new2;
    set new2;
    by usubjid day %if &xover=N %then %do; wt %end; die;

    if last.day;
run;

proc sort data=new2;
    by rx day;
run;

proc univariate data=new2 noprint;

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        var wt wtdie;
        by rx day;
        output out=main_plt sum=sumWgts wgtDead;
run;

data finalkm;
    set main_plt;
    by rx day;
    retain kmw;

    if first.rx and first.day then kmw=1;
    else do;

        if kmw ne 0 and sumWgts ne 0 then
            kmw=kmw*(1-wgtDead/sumWgts);
    end;
run;

data finalkm;
    set finalkm;
    month=day/30.24;

    %if &xover=Y %then %do;
    month=day/30.41;
    %end;
run;

data _finalkm;
    set finalkm;
    _rx=rx;
    run;

%if &xover=Y %then %do;

proc sql noprint;
    select probchisq into: pv_strat from &instratds;
    select strip(put(HazardRatio,5.3))||' ('||strip(put(HRLowerCL,5.3))||', '||strip(put(HRUpperCL,5.3))||')')
into: hr_strat from &instratds;

    %if &inunstratds^= %then %do;
    select strip(put(pvalue,pvalue6.4)) into: pv_unstrat from &inunstratds;

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%end;
quit;

proc format;
  value stratums      0='Placebo'
                     1='Active';

  value stratum       0='Placebo'
                     1='Active';

run;
options nodate;
title "Time to OS after Adjusting for Crossover related Confounding";
title2 "Using &method Approach";

ods exclude none;
ods graphics / attrpriority=none noscale tipmax=2500 imagefmt=png width=768px height=500px noborder;

proc sgplot data=_finalkm;
  format rx stratums. _rx stratum.;
  styleattrs datacontrastcolors=(blue red) datalinepatterns=(solid solid);
  step x=month y=kwmw / attrid=_rx group=_rx name='e' lineattrs=(thickness=2) nomissinggroup ;
  xaxis values=(0 to 80 by 5) label="Time to Death (Months)" labelattrs=(size=13pt weight=normal);
  yaxis values=(0 to 1 by .1) label="Adjusted Survival Probability" labelattrs=(size=13pt weight=normal);
  keylegend 'e' / exclude=("") sortorder=ascending location=inside valueattrs=(size=12pt weight=normal)
              down=2 noborder position=topright;
  refline .5 / lineattrs=(pattern=3);

  inset (      "From &method Adjusted Modeling: " = %trim(" ")
              "      Hazard Ratio: &hr_strat " = %trim(" ")
              "      P-value, Wt. Stratified Cox Model = &pv_strat " = %trim(" ")
              %if &inunstratds^= %then %do;
              "      P-value, Wt. Unstratified Log Rank Test = &pv_unstrat " = %trim(" ")
              %end;*/
  %if &method=MSM %then %do;
  "      Median: Active=68.3, Placebo=42.7 " = %trim(" ")
  %end;*/
  %if &method=IPCW %then %do;
  "      Median: Active=69.7, Placebo=44.1 " = %trim(" ")
  %end;

  )/ textattrs = (size=12 color=black) position=bottomleft;

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run;
ods graphics off;
%end;

%else %do;
PROC SQL;
    CREATE TABLE getmedian AS
    SELECT rx , kmw , 'month'n
    FROM WORK.FINALKM WHERE 0.47<kmw<=0.5
    order by rx, kmw descending, month descending;
QUIT;

data getmedian;
    set getmedian;
    by rx descending kmw descending month ;
    if first.rx;
run;

proc sql noprint;
    select probchisq into: pv_strat from &instratds;
    select strip(put(HazardRatio,5.3)||' ('||strip(put(HRLowerCL,5.3))||', '||strip(put(HRUpperCL,5.3))||')')
into: hr_strat from &instratds;
    %if &inunstratds ^= %then %do;
    select strip(put(pvalue,pvalue6.4)) into: pv_unstrat from &inunstratds;
    %end;

    select strip(put(month,5.1)) into: med_p from getmedian where rx=0;
    select strip(put(month,5.1))||',' into: med_o from getmedian where rx=1;
quit;

proc format;
    value stratums      0='Placebo'
                      1='Orterone';

    value stratum       0='Placebo'
                      1='Orterone';

run;

options nodate;
title "Time to OS after Adjusting for Alt. Rx effect";

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%if &method=TSE %then %do;
title "Time to OS after Adjusting for RCTs with Two-way Switching";
%end;
title2 "Using &method Approach";

ods exclude none;
ods graphics / attrpriority=none noscale tipmax=2500 imagefmt=png width=768px height=500px noborder;

proc sgplot data=_finalkm;
  format rx stratums. _rx stratum.;
  styleattrs datacontrastcolors=(blue red) datalinepatterns=(solid solid);
  step x=month y=kmw / attrid=_rx group=_rx name='e' lineattrs=(thickness=2) nomissinggroup ;
  xaxis values=(0 to 40 by 5) label="Time to Death (Months)" labelattrs=(size=13pt weight=normal);
  yaxis values=(0 to 1 by .1) label="Adjusted Survival Probability" labelattrs=(size=13pt weight=normal);
  keylegend 'e' / exclude=("") sortorder=ascending location=inside valueattrs=(size=12pt weight=normal)
               down=2 noborder position=topright;
  refline .5 / lineattrs=(pattern=3);

  inset (      "From &method Adjusted Modeling: " = %trim(" ")
             "      Hazard Ratio: &hr_strat " = %trim(" ")
             "      P-value, Wt. Stratified Cox Model = &pv_strat " = %trim(" ")
             %if &method=IPCW or &method=MSM %then %do;
             "      P-value, Wt. Unstratified Log Rank Test = &pv_unstrat " = %trim(" ")
             %end;
/* %if &method=MSM %then %do;
   "      Median: Orterone=34.6, Placebo=27.1 " = %trim(" ")
   %end;*/
%if &method=IPCW %then %do;
  "      Median: Orterone=30.1, Placebo=27.1 " = %trim(" ")
  %end;

%else %if &method=MSM %then %do;
  "      Median: Orterone=&med_o Placebo=&med_p " = %trim(" ")
  %end;

%else %if &method=TSE %then %do;
  "      Median: Orterone=35.3 Placebo=29.9 " = %trim(" ")
  %end;
)

  / textattrs = (size=12 color=black) position=bottomleft;

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run;  
ods graphics off;  
%end;  
  
%mend macro_km_plot;
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